

# Which original divisor words survive an Erdős–Straus trace descent?

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## Abstract

For positive rational ES tails with integral sum, we determine exactly when replacing the residual by a divisor preserves the original integer word and produces an integral solution. Such a descent is guaranteed for all inputs precisely for the nine words dividing 36. For each other word we construct infinitely many hard-prime inputs for which every such return, including the middle complement, fails. The primes still have explicit endpoint solutions. Initial source occupancy at an arbitrary prime is not asserted.

Here “hard prime” means  $p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}$ .

Let  $p \equiv 1 \pmod{4}$  be prime,  $p/4 < a < p/2$  an integer,  $R = 4a - p$ , and  $u \mid a^2$ . Retain the E/M channel label and put

$$G_E = 4u + 1, \quad G_M = p + 4u, \quad K(n) = \prod_{l^e \parallel n} l^{\lceil e/2 \rceil}, \quad m = 4K(u).$$

The positive rational triples

$$E : \left( a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left( a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right)$$

satisfy ES. Their tail sum is an integer exactly when  $R \mid G_c^2$ ; the common reduced denominator is  $d = R/\gcd(R, G_c)$  and  $d^2 \mid R$ . Indeed their sums are  $(a + pu)^2/(Ru)$  and  $p(a + u)^2/(Ru)$ , respectively. We consider proper trace sources, where  $d > 1$ . The word  $u$  is an available integer divisor, but the full residual gate is not yet satisfied. These are the classical Type-I/II coordinates before the full gate, not an alternative definition of ES [1, Propositions 2.2 and 2.6].

**Theorem 1** (The complete original return). *Every same-channel return keeping  $p, u$  and replacing  $R$  by a divisor is indexed by exactly*

$$\boxed{k \in \mathbb{Z}_{>0}, \quad k \mid R, \quad d \mid k, \quad k \equiv 1 \pmod{4K(u)}}.$$

It gives  $R' = R/k$ ,  $a' = (p + R')/4$ ,  $u' = u$  and  $a' < a$ .

*Proof.* The full target gate  $R/k \mid G_c$  is equivalent to  $d \mid k$ . The target integer denominator and word budget together are  $p + R/k \equiv 0 \pmod{4K(u)}$ . The input gives  $p + R \equiv 0$  modulo the same modulus, and  $p$  is a unit there. Multiplication by  $k$  proves the asserted congruence equivalence. It also gives  $k \equiv 1 \pmod{4}$ , hence  $R' \equiv 3 \pmod{4}$ ,  $3 \leq R' < R < p$  and the first-half range.

For a literal channel-labelled output put  $g = \gcd(a', u)$ ,  $(h, r, s) = (g^2/u, u/g, a'/g)$ . Prime valuations give positive integers with  $a' = hrs$ ,  $u = hr^2$ ,  $\gcd(r, s) = 1$ . For E use  $\kappa = (pr + s)/R'$  and  $(a', hsk, phr\kappa)$ ; for M use  $\lambda = (r + s)/R'$  and  $(a', phs\lambda, phr\lambda)$ . The gates prove integrality, and multiplication verifies the identity. The respective labelled inverses are  $a'^2/(R'y - pa')$  and  $pa'^2/(R'y - pa')$ .  $\square$

Here and below  $k$  is a positive integer. The full inverse fibre above a fixed original target satisfying  $R' = 4a' - p$ ,  $3 \leq R' < p$ ,  $R' \equiv 3 \pmod{4}$ ,  $u \mid (a')^2$ , and  $R' \mid G_c$  retains all  $1 \leq k \leq (p - 1)/R'$  with  $k \equiv 1 \pmod{4K(u)}$ ,  $R'k \mid G_c^2$ , and  $R'k \nmid G_c$ . Set  $R = R'k$  and  $a = (p + R)/4$ . Thus coefficient aggregation is over known finite fibres, not over unmarked residue support.

Write  $R = \delta t^2$  with  $\delta$  its squarefree part, and  $q_0 = K(d)$ . Then  $q_0 \mid t$ . The square deletions are exactly

$$k = q^2, \quad q_0 \mid q \mid t, \quad q^2 \equiv 1 \pmod{4K(u)}.$$

**Theorem 2** (Exact universal seed list). *For a fixed positive integer  $u$ , every proper trace source at every prime  $p \equiv 1 \pmod{4}$  has the preceding return if and only if*

$$u \in \{1, 2, 3, 4, 6, 9, 12, 18, 36\}.$$

*For each listed word,  $q = q_0$  gives a one-step return. Its complete square-deletion family has  $\tau(t/q_0)$  distinct original targets.*

*Proof.* A modulus  $m$  divisible by four has every unit square equal to one exactly when  $m \mid 24$ . To check this directly, an odd prime power has exactly two solutions to  $x^2 = 1$ ; modulus four has two and  $2^\alpha$ ,  $\alpha \geq 3$ , has four. Counting the kernel of the square map proves that its image has order

$$2^{\max(0, \alpha-3)} \prod_{i=1}^r l_i^{e_i-1} (l_i - 1)/2,$$

for  $m = 2^\alpha \prod_{i=1}^r l_i^{e_i}$  with distinct odd  $l_i$ . It is one precisely for  $m = 4, 8, 12, 24$ , equivalently  $K(u) \mid 6$  or  $u \mid 36$ . All available  $q$  are units modulo  $m$ . Thus for these words every  $q_0 \mid q \mid t$  works, giving the stated count and strict descent.

For necessity let  $m = 4K(u) \nmid 24$ . Choose a unit  $b \pmod{m}$  with  $b^2 \neq 1$ ; replace its sign to arrange  $b \equiv 1 \pmod{4}$ . Dirichlet's theorem [2, Theorem 18.1] gives distinct primes

$$\delta \equiv -b^2 \pmod{m}, \quad t \equiv b^{-1} \pmod{m}, \quad \delta, t > \max(u, 7).$$

Thus  $\delta \equiv 3 \pmod{4}$ ,  $t \equiv 1 \pmod{4}$  and both are coprime to  $L = \text{lcm}(840, m)$ . Set  $R = \delta t^2$ , and take a prime  $e \equiv 3 \pmod{4}$  coprime to  $LR$ . The reduced CRT class

$$p \equiv 1 \pmod{L}, \quad p \equiv \delta t - 4u \pmod{R}, \quad p \equiv -1 \pmod{e}$$

contains infinitely many primes. Take  $p > \max(R, e, B(u)R)$ , where  $B(u) = \prod_l l^{\lfloor v_l(u)/2 \rfloor}$ . Since  $840 \mid L$ , all these primes lie in the hard class  $1 \pmod{840}$ . Then  $a = (p+R)/4$  is an integer multiple of  $K(u)$ , since  $R \equiv -1 \pmod{m}$ . The original M word  $u \mid a^2$  has exact denominator  $d = t > 1$ .

The only candidate target residuals dividing  $R$ , dividing  $p+4u$ , and congruent to three modulo four are  $\delta, \delta t, R$ . The last fails the gate. The first two fail the word budget:  $p + \delta \equiv 1 - b^2 \neq 0$  and  $p + \delta t \equiv 1 - b \neq 0$  modulo  $m$ . Even the other middle orientation cannot escape: for  $u^* = a^2/u$ ,  $4K(u^*) = (p+R)/B(u) > R$ . Moreover  $4u(p+4u^*) \equiv p(p+4u) \pmod{R}$ , so its exact denominator remains  $t > 1$ . Thus no allowed deletion factor can be one modulo its budget.

These are not ES counterexamples. Put  $c = (p+1)/e$  and  $j = (e+1)/4$ . The original middle state  $(h, r, s, \lambda) = (j, 1, c, 1)$  returns  $(jc, pj, c, pj)$ . Its first-half range follows from  $c(e-1) > 2$ . This supplies a genuine solution at every prime used in the obstruction.  $\square$

**Why full multiplicities and non-square deletions matter.** At  $p = 1009, a = 321, R = 275, u = 9$ , the proper M trace has tails  $215926/5, 6054/5$ . The old full E and M gates are both empty, but  $k = 25$  returns

$$\frac{4}{1009} = \frac{1}{255} + \frac{1}{686120} + \frac{1}{24216}.$$

At  $p = 51361, a = 14845, R = 8019, u = 5$ , the sole deletion is  $k = 81$  and the new residual is 99, still containing a square. Stripping every square instead loses the original word. At  $p = 1108801, a = 279295, R = 8379, u = 5$ , the complete deletion set is  $\{21, 441\}$ : an additional available factor seven repairs the residue of the forced factor three. Its integer coefficient is two, not an absent coefficient after reduction modulo two; the two labelled representatives are  $v = 7, 147$ , while the retained original word is the single word  $u = 5$ .

The complete proof and executable certificates retain the inverse fibres and original coefficient counts. A trace source is not guaranteed at every prime; universal ES occupancy and historical priority are not claimed.

## The exact nine-word boundary—and what fails outside it

A uniform one-step square deletion exists iff  $H_{4K(u)} = \{1\}$  iff  $u \mid 36$ .

### All and only the automatic words

|                                                 |                                                 |                                                 |
|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| $u = 1$<br>$K(u) = 1, m = 4$<br>$H_m = \{1\}$   | $u = 2$<br>$K(u) = 2, m = 8$<br>$H_m = \{1\}$   | $u = 3$<br>$K(u) = 3, m = 12$<br>$H_m = \{1\}$  |
| $u = 4$<br>$K(u) = 2, m = 8$<br>$H_m = \{1\}$   | $u = 6$<br>$K(u) = 6, m = 24$<br>$H_m = \{1\}$  | $u = 9$<br>$K(u) = 3, m = 12$<br>$H_m = \{1\}$  |
| $u = 12$<br>$K(u) = 6, m = 24$<br>$H_m = \{1\}$ | $u = 18$<br>$K(u) = 6, m = 24$<br>$H_m = \{1\}$ | $u = 36$<br>$K(u) = 6, m = 24$<br>$H_m = \{1\}$ |

For every proper source at these words,  $q_0 = K(d)$  gives  $k = q_0^2$  and  $R' = R/q_0^2$ .

The complete square family has  $\tau(t/q_0)$  distinct targets; no source-existence claim is added.

### Outside boundary: the exact $u = 8$ obstruction

$$m = 16, \quad H_{16} = \{1, 9\}, \quad b = 5 \\ \delta = 71, t = 13, R = \delta t^2 = 11999 \\ p = 48116881, a = 12032220, d = t = 13$$

Every possible same-word target residual fails:

$$R' = \delta = 71 \\ p + R' \equiv 8 \equiv 0 \pmod{16}$$

$$R' = \delta t = 923 \\ p + R' \equiv 12 \equiv 0 \pmod{16}$$

middle complement  $u^* = a^2/u$   
 $4K(u^*) = 24064440 > R$ ; its exact denominator is still 13

**This blocks the residual-divisor method, not Erdős–Straus.**

A separate endpoint state (13122786, 631427532350466, 144350643)  
has reciprocal sum  $4/48116881$ .

Figure 1: The exact universal nine-word boundary and the explicit  $u = 8$  obstruction to the residual-divisor method. The obstruction prime has a separately displayed integer ES endpoint and is not an ES counterexample.

## References

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Austral. Math. Soc.*, 94, 50–105. arXiv:1107.1010, §2, Proposition 2.2 and Proposition 2.6.
- [2] Sutherland, A. V. (2017). *18.785 Lecture 18: Dirichlet L-functions, primes in arithmetic progressions*. MIT, Theorem 18.1.